

Pure Death Process

A pure death process $N(t)$ starts from an initial population $i > 0$ and eventually reduces to zero since individuals or items 'die' at a certain rate. There is no 'birth' in the population. Let μ_n be the death rate when the population size is n .

Conditions:

In a very small time-interval $(t, t + h]$:

$$P(\text{one death}) = \mu_n h + o(h)$$

$$P(\text{no death}) = 1 - \mu_n h + o(h)$$

$$P(\text{more than one death}) = o(h).$$

Differential Equations:

By comparing with the pure birth process, we have

$$p'_n(t) = -\mu_n p_n(t) + \mu_{n+1} p_{n+1}(t)$$

$$p'_i(t) = -\mu_i p_i(t)$$

When $\mu_n = n\mu$, the differential equations are given by:

$$p'_n(t) = -n\mu p_n(t) + (n+1)\mu p_{n+1}(t)$$

$$p'_i(t) = -i\mu p_i(t)$$

It can be shown that

$$p_n(t) = \binom{i}{n} e^{-n\mu t} (1 - e^{-\mu t})^{i-n}; \quad n = 0, 1, 2, \dots, i.$$

which is a binomial distribution with parameters i and $p = e^{-\mu t}$ (probability of survival of each individual or item).

The mean is obtained as

$$E(N(t)) = ie^{-\mu t}.$$

Exercise:

In a store, there are 20 packets of candy for sale. The sale rate depends on the number of packets remaining. For one packet, the rate is 0.5/hour.

(a) What is the probability that after 3 hours 5 packets will be remaining?

- (b) What is the probability that all the packets will be sold out in an 8-hour day?
(c) What is the average number of items sold in 4 hours?

Solution:

This is a pure death process with $i = 20$, $\mu = 0.5/\text{hour}$.

(a)

$$\begin{aligned} p_5(3) &= \binom{20}{5} e^{-5 \times 0.5 \times 3} (1 - e^{-0.5 \times 3})^{20-5} \\ &= 0.1943 \end{aligned}$$

(b)

$$\begin{aligned} p_0(3) &= \binom{20}{0} e^{-0 \times 0.5 \times 3} (1 - e^{-0.5 \times 3})^{20-0} \\ &= 0.6909 \end{aligned}$$

(c)

Average number of items remaining:

$$E(N(4)) = 20 e^{-0.5 \times 4} = 2.71$$

Therefore, average number of items sold:

$$20 - 2.71 = 17.29$$

Birth-Death Process

A birth-death process $N(t)$ starts with an initial population $i \geq 0$. When the population size is n , births occur with rate λ_n and deaths occur with rate μ_n .

Conditions

A. Conditions related to birth

In a very small time-interval $(t, t + h]$:

$$P(\text{one birth}) = \lambda_n h + o(h)$$

$$P(\text{no birth}) = 1 - \lambda_n h + o(h)$$

$$P(\text{more than one birth}) = o(h).$$

B. Conditions related to death

In a very small time-interval $(t, t + h]$:

$$P(\text{one death}) = \mu_n h + o(h)$$

$$P(\text{no death}) = 1 - \mu_n h + o(h)$$

$$P(\text{more than one death}) = o(h).$$

C. Condition of independence

Birth and death occur independently of each other.

Differential Equations

Combining pure birth process and pure death process, we have

$$p'_n(t) = -(\lambda_n + \mu_n) p_n(t) + \lambda_{n-1} p_{n-1}(t) + \mu_{n+1} p_{n+1}(t)$$

Initial conditions: $p_i(0) = 1$ and $p_n(0) = 0$ if $n \neq i$.

$$p'_0(t) = \mu_1 p_1(t) - \lambda_0 p_0(t)$$

Steady-state solution

When the process runs for a long time ($t \rightarrow \infty$), the differential equations take the form:

$$\lambda_{n-1} p_{n-1} + \mu_{n+1} p_{n+1} - (\lambda_n + \mu_n) p_n = 0$$

$$\mu_1 p_1 - \lambda_0 p_0 = 0$$

The solutions are:

$$p_n = \frac{\lambda_0 \lambda_1 \cdots \lambda_{n-1}}{\mu_1 \mu_2 \cdots \mu_n} p_0$$

$$p_0 = \left(1 + \sum_{n \geq 1} \frac{\lambda_0 \lambda_1 \cdots \lambda_{n-1}}{\mu_1 \mu_2 \cdots \mu_n} \right)^{-1}$$

Solution exists if

$$\sum_{n \geq 1} \frac{\lambda_0 \lambda_1 \cdots \lambda_{n-1}}{\mu_1 \mu_2 \cdots \mu_n} < \infty.$$

Examples

1. Single server queue: $\lambda_n = \lambda, \mu_n = \mu$.
A small branch of bank with only one teller.
2. Infinite server queue: $\lambda_n = \lambda, \mu_n = n\mu$.
Here, queue length will be zero, as in a food party with lots of packets.
3. Infinite length queue: $\lambda_n = n\lambda, \mu_n = \mu$.
A government office where there is only one server, but there is a sudden craze in people to arrive there.
4. Linear growth model: $\lambda_n = n\lambda, \mu_n = n\mu$.
A reconciliation party where people of one village go to another village.

Queueing process

A process that consists of a servicing facility, arrival of customers for getting service and departure of customers after getting service is called a queueing process. It has the following four components:

- a) *Input process*: Arrivals are controlled by factors external to the system and is represented by random variables.
- b) *Service mechanism*: Number of servers is usually not random. Number of customers getting service within a period of time is random.
- c) *Queue discipline*: Rules such as 'first-come, first-served', 'last-come, first-served', 'random selection for service', 'priority service' etc. are employed.
- d) *Number of servers*: It can be one or two or, in some cases, infinite.
- e) *Number of queues*: Sometimes there are two or more queues. For example, males and females wait in different queues in Bangladesh.

Description of a queueing system

A queueing system can be described by specifying the input (arrival) distribution, service time distribution and the number of servers. Queue discipline is rather qualitative and should be described in the proper context. Number of queues will be one in our discussion. Thus, we write

Input distribution / Service time distribution / Number of servers

Some standard notation:

M : Poisson distribution (for input), exponential distribution (for service)

GI : General distribution (for input)

G : General distribution (for service)

D : Constant length of time (service)

Example

a) $M/M/1$

Input distribution is Poisson, service time distribution is exponential, and number of servers is one.

b) $M/G/2$

Input distribution is Poisson, service time distribution is general, and number of servers is two.

c) $M/D/1$

Input distribution is Poisson, service time is constant, and number of servers is one.

Variables of interest

$Q(t)$ = Queue length (number of customers waiting or being served) at time t

$W(t)$ = Waiting time (amount of time a customer arriving at time t has to wait)

$S(t)$ = System time (amount of time a customer has to stay in the system)

 $M/M/1$ Queue

Arrival distribution: Poisson with rate λ

Service time distribution: Exponential with rate μ

Traffic intensity of the process is defined as:

$$\rho = \frac{\lambda}{\mu}$$

Limiting distribution (steady state distribution) of the process exists if $\rho < 1$.

Distribution of queue length

Let $Q(\infty) = Q$ be the number of customers in the system (waiting or being served) in steady state. It can be shown that the distribution of Q is given by

$$\pi_n = P(Q = n) = \rho^n(1 - \rho), \quad n = 0, 1, 2, \dots$$

which is a geometric distribution with probability of “success” $(1 - \rho)$.

$$E(Q) = \frac{\rho}{1 - \rho}$$

$$V(Q) = \frac{\rho}{(1 - \rho)^2}$$

Distribution of system time

Let $S(\infty) = S$ be amount of time a newcomer has to stay in the system (including his individual service time) in steady state. The newcomer finds that there are Q customers already in the system. The conditional distribution of S , given $Q = n$, is gamma with parameters $(n + 1)$ and μ , since it is the sum of $(n + 1)$ service times each following exponential with parameter μ . That is,

$$S | n \sim \text{gamma}(n + 1, \mu)$$

It can be shown that the unconditional distribution of S is exponential with parameter $\mu(1 - \rho)$. That is,

$$S \sim \text{exponential}(\mu(1 - \rho))$$

$$E(S) = \frac{1}{\mu(1 - \rho)}$$

$$V(S) = \frac{1}{\mu^2(1 - \rho)^2}$$

Mean waiting time

Let $W(\infty) = W$ be the amount of time a newcomer arriving at time t has to wait before his service starts. Since $W = S - V$ where V is the newcomer’s individual service time, we have

$$\begin{aligned}
 E(W) &= E(S) - E(V) \\
 &= \frac{1}{\mu(1-\rho)} - \frac{1}{\mu} \\
 &= \frac{\rho}{\mu(1-\rho)}
 \end{aligned}$$

Example

Arrival at a telephone booth forms a Poisson process with mean 12/hour. Length of a telephone call follows exponential with mean 2 minutes.

- What is the probability that, on arrival, a person will find the telephone occupied?
- What is the average length of queue?
- What is the average waiting time?
- What is the probability that a newcomer has to stay in the system for more than 2 minutes?

Solution

It is an $M/M/1$ queue with $\lambda = \frac{1}{5}$ /minute and $\mu = \frac{1}{2}$ /minutes.

a)

$$\rho = \frac{\lambda}{\mu} = \frac{1/5}{1/2} = 0.4$$

$$\begin{aligned}
 P(Q > 0) &= 1 - P(Q = 0) \\
 &= 1 - \rho^0(1 - \rho) \\
 &= \rho \\
 &= 0.4
 \end{aligned}$$

b)

$$E(Q) = \frac{\rho}{1 - \rho} = 0.67$$

c)

$$E(W) = \frac{\rho}{\mu(1 - \rho)} = 1.33 \text{ minutes}$$

d)

$$S \sim \text{exponential}(\mu(1 - \rho))$$

$$P(S > x) = \exp(-\mu(1 - \rho)x) = \exp(-0.5(1 - 0.4)2) = 0.549$$

M/M/s Queue

Arrival distribution: Poisson with rate λ

Service time distribution: Exponential with rate μ

There are s servers in the system. Arriving customers queue up in a single line. As soon as a server becomes free, the customer at the front of the queue is taken for service. Here,

$$\lambda_n = \lambda$$

$$\mu_n = \begin{cases} s\mu, & n \geq s \\ n\mu, & n < s \end{cases}$$

Traffic intensity of the system is defined as:

$$\rho = \frac{\lambda}{s\mu}$$

Limiting distribution (steady state distribution) of the process exists if $\rho < 1$.

Limiting distribution of Q

$$\pi_0 = P(Q = 0)$$

$$= \left(\sum_{i=0}^s \frac{(s\rho)^i}{i!} + \frac{s^s \rho^{s+1}}{s!} (1 - \rho)^{-1} \right)^{-1}$$

$$\pi_n = \begin{cases} \frac{(s\rho)^n}{n!} \pi_0, & n \leq s \\ \frac{s^s \rho^n}{s!} \pi_0, & n > s \end{cases}$$

Note that $\pi_n = \rho^{n-s} \pi_s, n \geq s$.

$$E(Q) = s\rho + \frac{\rho \pi_s}{(1 - \rho)^2}$$

Average waiting time:

$$E(W) = \frac{\pi_s}{s \mu (1 - \rho)^2}$$

Example

A supermarket has 3 girls ringing up sales at the counters. People arrive in a Poisson process with a rate of 30/hour. Service time follows exponential distribution with mean 5 minutes.

- (a) Under steady state condition, what is the probability that all the girls will be busy?
- (b) What is the average number of customers in the system?
- (c) What is the average waiting time of a customer who has just arrived?

Solution

It is an $M/M/s$ queue with $s = 3$, $\lambda = \frac{1}{2}$ /minute and $\mu = \frac{1}{5}$ / minutes.

$$\rho = \frac{\lambda}{s \mu} = \frac{0.5}{3(0.2)} = \frac{5}{6}$$

$$\pi_0 = \left(\sum_{i=0}^s \frac{(s\rho)^i}{i!} + \frac{s^s \rho^{s+1}}{s!} (1 - \rho)^{-1} \right)^{-1} = 0.0449$$

$$\pi_1 = \frac{(s\rho)^1}{1!} \pi_0 = 0.1123$$

$$\pi_2 = \frac{(s\rho)^2}{2!} \pi_0 = 0.1404$$

$$\pi_3 = \frac{(s\rho)^3}{3!} \pi_0 = 0.1170$$

a)

$$1 - \pi_0 - \pi_1 - \pi_2 = 0.7022$$

b)

$$E(Q) = s\rho + \frac{\rho \pi_3}{(1 - \rho)^2} = 6.0$$

c)

$$E(W) = \frac{\pi_3}{s \mu (1 - \rho)^2} = 7.0$$

The End